

PROGRAM 25

AIM

To implement a Prolog program to identify the colour of a fruit using backtracking.

ALGORITHM

1. Start the program.
2. Define facts containing fruits and their colours.
3. Create a rule to find the colour of a given fruit.
4. Enter a query for a fruit.
5. Prolog searches the facts and returns the matching colour.
6. If multiple matching facts exist, Prolog uses backtracking to find the next possible answer.
7. Display the fruit and its corresponding colour.
8. Stop the program.

PROGRAM

```
fruit_color(apple, red).  
fruit_color(banana, yellow).  
fruit_color(orange, orange).  
fruit_color(grapes, green).  
fruit_color(mango, yellow).  
fruit_color(guava, green).  
find_color(Fruit, Color) :-  
    fruit_color(Fruit, Color).
```

OUTPUT

```
41 ?- consult(['c:/Users/Hema Neepe/OneDrive/Documents/Friut_and_its_color.pl']).  
true.  
  
42 ?- find_color(apple, Color).  
Color = red.  
  
43 ?- find_color(banana, Color).  
Color = yellow.  
  
44 ?- find_color(mango, Color).  
Color = yellow.
```

RESULT

The Prolog program was successfully implemented to identify the colour of a fruit, and backtracking was used by Prolog to search for matching facts.